

Intelligent Behavior Analysis and Therapy Support System for Autism



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System for Autism**

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The thesis is to be submitted to the Department of Computer Science,
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the requirement for Bachelor of Science in Computer Science.

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Declaration

I declare that the work contained in this thesis is my own, except where explicitly stated otherwise. In addition this work has not been submitted to obtain another degree or professional qualification.

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Acknowledgments

The acknowledgements and the people to thank go here, don't forget to include your project advisor. . .

For/Dedicated to/To my...

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Abbreviations

LAH List Abbreviations **Here**

Abstract

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Chapter 1

Introduction

Autism Spectrum Disorder (ASD) is a neuro-developmental condition characterized by challenges and differences in social communication and interaction, repetitive behaviors, or focused interests. The condition shows symptoms in the early life of a human and then persists throughout their lifetime. ASD continues to increase globally, with approximately 1 in 36 children affected by this disorder. Effective intervention for autistic children relies heavily on early identification, behavioral analysis, and personalized therapy that adapts to each individual's unique characteristics. Therapists and caregivers collect large amounts of observations in common clinical practices. However, most of their observations remain under-utilized due to manual recordings and subjective analysis.

The machine learning components draw upon supervised learning techniques, time-series analysis, and pattern recognition algorithms that have demonstrated efficacy in behavioral prediction and classification tasks. Meanwhile, the user interface design incorporates principles of human-computer interaction specifically adapted for clinical workflows, ensuring that powerful analytical capabilities remain accessible to users with varying levels of technical expertise. There is an urgent need for technological intervention in the behavioral clinical healthcare sector to improve the efficiency, precision, and management of therapy for individuals with Autism Spectrum Disorder (ASD). Traditionally, behavioral data analysis has been based on subjective observations and low-frequency data collection. The integration of Artificial Intelligence and data-driven platforms serves as a catalyst in transforming this experience for therapists and caregivers. Many existing solutions fail to provide automated, intelligent pattern recognition or personalized, evidence-backed therapy recommendations.

This Final Year Project presents the design and development of an *Intelligent*

Behavior Analysis and Therapy Support System for Autism, a secure and data-driven platform to support therapists and caregivers in monitoring behavior, analyzing patterns, and generating personalized therapy recommendations. The system is developed as a decision-support tool rather than a replacement for professional advice, using machine learning for pattern recognition and Retrieval-Augmented generation (RAG) framework for therapy suggestions to improve the quality and efficiency of behavioral therapy suggestions.

1.1 Background

Autism Spectrum Disorder (ASD) is a complex neuro-developmental disorder that is categorized by challenges and differences in social communication, behavior repetition. The heterogeneous nature of ASD means that each affected individual presents unique characteristics and behavioral patterns, which reflects the need for personalized behavioral therapy suggestions. There are operational limitations in current clinical practices. Therapists and caregivers typically document behavior observations over a long period of time in different sessions and settings. This nature of data collection introduces inconsistencies in recording details and accuracy. The volume of behavioral data generated over a period of time in different settings and environments generates a sheer volume of behavioral data creating an analytical burden for the therapist to recommend a therapy for the patient. This process is long and prone to human error.

Advancements in the digital health technology and artificial intelligence has opened new doors for addressing these challenges. Machine learning algorithms excel in identifying complex patterns, and the latest web technologies enable secure data management. Explainable AI (XAI) has also introduced a new paradigm especially in the healthcare. XAI has become a critical part of machine learning systems used in healthcare. This project combines these technologies to create a comprehensive system tailored to the requirements of personalized autism therapy.

The technological foundation of this system is based on established frameworks and methodologies proven to be effective in the healthcare applications. The system addresses the security requirements for handling sensitive healthcare data as mandated by the regulations such as HIPAA and GDPR by employing robust authentication mechanisms.

1.2 Problem Statement

Therapists and caregivers face ongoing challenges in managing the high volume of behavioral data required for effective ASD intervention. Manual processes for

data logging and pattern identification are common, leading to inefficiencies and the risk of subjective bias.

Most current systems do not offer automated video analysis or context-aware therapy suggestions. This lack of intelligent support puts unnecessary pressure on clinical professionals, limiting their ability to focus on direct patient care. Hence, there is a critical need for a platform that leverages Long Short-Term Memory (LSTM) networks and Large Language Models (LLMs) to enhance the efficiency and personalization of autism therapy.

1.3 Motivation

The motivation of the project is the technological opportunity and the need of the society for this system. The project offers an opportunity to apply modern software engineering principles, machine learning, and secure system design to a real-world healthcare-related problem. The challenge of translating complex behavioral data into clear, actionable recommendations aligns strongly with current research trends in artificial intelligence and digital therapeutics.

Improving the lives of autistic children by improving the quality of therapy suggestions can have a great impact on society. Even a small improvement in the therapy suggestion can have a great impact on a life.

1.4 Scope

The scope of the project is detailed below:

- **Secure Multi-Role Authentication:** Distinct access for Therapists and Caregivers.
- **Structured Data Logging:** Mobile-optimized interface for recording behavioral incidents.
- **Standardized Video Collection:** Guided protocols for 11 physical actions (e.g., Arm swing, Frog Pose).
- **Intelligent Pattern Analysis:** Automated behavior recognition using Machine Learning.
- **Personalized Therapy Support:** RAG framework for data-driven intervention suggestions.
- **Security:** The system will secure the sensitive data according to the HIPPA and GDPR regulations.

The following details are the out of scope of the project:

- Direct interaction with medical or diagnostic hardware
- Prescription management or medication tracking
- Replacement of professional therapists or clinicians

The system is explicitly positioned as a decision-support and analysis tool rather than a clinical authority.

1.5 Objectives

The core objectives of the system are as follows:

- Automate behavioral analysis to reduce the manual workload of therapists.
- Design a secure, scalable, and modular software architecture suitable for handling sensitive behavioral data
- Develop a user-friendly platform that supports efficient behavioral data collection
- Apply machine learning techniques to identify meaningful behavioral patterns and trends
- Generate personalized therapy support recommendations based on data-driven insights
- Provide explainable reports that offer confidence scores and clinical reasoning for AI predictions.
- Ensure compliance with data security and privacy best practices
- Evaluate the effectiveness of the system as a support tool for therapists and caregivers

1.6 Challenges

The development process involves several technical and clinical challenges:

- Ensuring high accuracy in detecting nuanced behavioral patterns across different patients.
- Maintaining PHI security during the transit and storage of large video files.

- Developing a low-latency API for real-time ML model serving.
- Designing and developing machine learning model that provides interpretable and clinically meaningful outputs
- Implementing a RAG pipeline that retrieves only the most relevant and clinically sound documents.
- Designing an intuitive UI that caregivers can use during active behavioral incidents without distraction.

1.7 Learning Outcomes

The following key learnings are expected upon the completion Of this project:

- Enhanced understanding of software requirement specification and system design methodologies
- Practical experience in secure web application development
- Applied knowledge of machine learning techniques for real-world data analysis
- Deeper insight into ethical, privacy, and security considerations in healthcare-related software systems
- Strengthened skills in technical documentation and project evaluation
- Improved ability to design scalable and modular system architectures

1.8 End Product

The final platform is a fully functional prototype of an AI-powered therapy support system that provides:

- Real-time behavioral logging and video upload capabilities.
- Automated and explainable analysis of standardized physical movements.
- Consolidated clinical reports identifying cross-video patterns.
- Interactive dashboards for therapists to review AI findings and manage patient goals.
- Secure PDF export of analytical findings and therapy suggestions.

- Comprehensive documentation, including system design, implementation details, and evaluation results

1.9 Completeness Criteria

The following deliverables define the successful completion of the project:

TABLE 1.1: Project Completeness Criteria

Deliverable	Description
Authentication & Profile Modules	Secure login and patient management.
Video Processing Pipeline	Successful upload of standardized videos based on 11 specified actions.
LSTM Analysis Engine	Functional ML model identifying patterns.
RAG Recommendation System	LLM-generated therapy suggestions.
Consent & Audit Module	Digital consent management and PHI audit trails.
Technical Documentation	Completed SRS, ERD, and Testing Reports.

1.10 Business Goals

The project aligns with long-term business and industry goals in the digital health and assistive technology sectors. The system has long-term potential as a commercial healthcare solution:

- **SaaS offering** for specialized autism clinics and therapy centers.
- **Standardized data platform** for researchers to study the effectiveness of long-term interventions.
- **Premium features** including real-time clinical alerts and advanced predictive analytics for patient regression.

Chapter 2

Background Study

2.1 Background Study

There has been significant progress in the application of artificial intelligence for supporting autism diagnosis, behavioral monitoring, and therapy suggestions. Multiple systems have explored video, audio, questionnaires, neuroimaging, and wearable sensors to detect ASD patterns and support clinical decision-making.

For early detection of autism, Khan et al. [4] introduced MCBERT, which fuses brain MRI features with BERT-based metadata embeddings to reach over 93% accuracy on the ABIDE dataset. Mahmood et al. [7] employed ResNet and Vision Transformer models for facial expression analysis with 91% accuracy, while Atlam et al. [1] achieved 98% accuracy using transfer-learning CNNs with explainability via LIME. Contreras et al. [2] further improved image-based classification by applying advanced preprocessing pipelines to pre-trained networks, boosting ViT accuracy to 92.7%.

Several systems have begun addressing behavioral monitoring for autistic patients. Zhu et al. [10] developed a machine learning system to quantify toddlers’ “response to name” behavior from video, achieving 92% agreement with human ratings. Shen et al. [9] advanced wearable-based monitoring with MIND, capable of forecasting restricted and repetitive behaviors (RRBs) with 92% accuracy seconds before onset. Chomistek et al. [3] integrated video, audio, and smart-glasses data in a composite AI framework to analyze ADOS-2 sessions, successfully detecting verbal and nonverbal requests with up to 94% F1 score. Kurian et al. [6] introduced m.AutNet, a personalized multimodal model achieving 88% accuracy for emotion classification in autistic children.

Therapy support technologies remain relatively underexplored compared to diagnostic systems. Kohli et al. [5] introduced a machine-learning recommendation framework for Applied Behavior Analysis (ABA) treatment planning, which achieved over 80% accuracy in predicting suitable therapy goals. However, most AI systems in the literature primarily emphasize detection and assessment, with limited integration of personalized intervention guidance.

Ranasingha et al. [8] presents an innovative deep learning framework that enhances autism severity classification by combining principles from physics with advanced sequence modeling. It proposes a Physics-Augmented Tuple Transformer (PATT) that encodes skeleton-based motion trajectories into a latent space and employs both physics-based and non-physics-based decoders to model motion dynamics. A classifier built on these embeddings predicts autism severity levels, leveraging motion features derived from datasets including MMASD, demonstrating state-of-the-art performance across multiple ASD severity recognition benchmarks and even generalizing to related tasks like fall prediction.

Although significant progress has been achieved in early screening, diagnosis, and behavioral monitoring, most current approaches treat detection and intervention as separate pipelines. Models are optimized either for high-accuracy classification of ASD-related behaviors or for retrospective analysis of clinical sessions. For instance, while systems like MIND [9] and m_AutNet [6] demonstrate strong capabilities in recognizing affective or repetitive behaviors, they do not translate these insights into actionable, context-aware therapy suggestions. This gap highlights the uniqueness of our system, which combines BAE for behavioral monitoring and TSE for therapy suggestion.

TABLE 2.1: Comparison of Related Systems

System	Datasets	Results	Features	Limitations
Khan et al. [4]	ABIDE-I dataset	Accuracy: 93.4%	CNN for neuroimaging	Limited to one dataset
Zhu et al. [10]	Video data from 125 toddlers	Accuracy: 74.8%	Response-to-Name behavior scoring	Single symptom only
Shen et al. [9]	Wearable sensor data	Accuracy: 73%	Detection & forecasting RRBs	No therapy support
Chomistek et al. [3]	Video/audio from ADOS-2 sessions	Accuracy: 98.2%	Detects verbal/non-verbal requests	No therapy support
Atlam et al. [1]	Kaggle facial images	Accuracy: 92.7%	Deep Learning	Detection only
Contreras et al. [2]	Kaggle facial images	Accuracy: 88%	Deep transfer learning	No therapy support
Kurian et al. [6]	Face + audio	Accuracy: 80%	Personalized emotional recognition	Emotion recognition only
Kohli et al. [5]	ABA treatment records	Accuracy: 92%	Recommendation system	No explainability
Mahmood et al. [7]	Facial images	Accuracy: 91%	ASD detection from faces	No therapy support
Ranasingha et al. [8]	MMASD	Accuracy: 95%	Transformer-based deep learning	No therapy support
Proposed System	2D / 3D Coordinates, EHR, and Logs	–	Integrated Behavior Analysis & therapy	–

Chapter 3

Requirements Specification

3.1 Introduction

3.1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of *Intelligent Behavior Analysis and Therapy Support System for Autism*. The purpose of this chapter is to present a description of the intended audience, behavior, quality attributes and constraints of the system. This chapter serves as the primary technical blueprint, specifying the detailed functional and non-functional features, the system's architecture, and the tech stack. It serves as a formal agreement between all stakeholders regarding what the system will do, how it will perform, and the conditions under which it will operate.

The primary objective of this system is to enhance the efficiency, consistency, and reliability of behavioral analysis by reducing manual data handling and subjective interpretation. The system is designed to assist therapists and caregivers in the structured collection, analysis, and interpretation of behavioral data for individuals with Autism Spectrum Disorder (ASD). The system aims to transform raw observational data into meaningful, data-driven insights that support personalized therapy suggestions using secure web technologies and machine learning-based behavioral pattern analysis with explainability reports.

It is a complete description of what the system will do, what behavior it will exhibit, constraints, and extensive details of how the system will behave, what services it will provide, and under what conditions it will operate. It is a valuable reference point for developers, designers, project supervisor, and evaluators

throughout the development process. It is intended to guide developers during implementation, support project supervisor and evaluators in assessing requirement fulfillment, and ensure that the final system aligns with its defined goals and scope.

3.1.2 Intended Audience

This document is primarily intended for the following groups:

- **Development Team:** To guide the implementation, coding, and testing process according to defined specifications.
- **Project Supervisor/Evaluators:** To assess the technical soundness, completeness and adherence of the final system to the initial requirements and design.
- **End Users (Therapists and Caregivers):** This document defines the features and usability standards that will ultimately benefit them through a more efficient and effective therapy tool.

3.2 Overall Description

3.2.1 Product Description

The Intelligent Behavior Analysis and Therapy Suggestion for Autism is a comprehensive web-based system that gathers behavioral videos of the autistic children, extracts key behavioral attributes through their 2D and 3D skeleton coordinates and transforms them into key actionable insights and therapy suggestion through machine learning and evidence-based decision support for therapists and caregivers.

The system is operated as a web application that can be accessible through desktop, mobile or any other device with the internet. The system is designed as a modular, tiered architecture comprising a front-end user interface, a backend containing the business logic of the application and a database for persistent storage. The system also have a separate AI service integrated with the backend for behavior analysis and therapy suggestion. The system is built to address the behavioral analysis and therapy suggestion management.

3.2.2 User Characteristics

The system is intended to support multiple user roles involved in autism therapy and system administration, each with distinct responsibilities and levels of technical expertise:

- **Administrators:** Responsible for general management of the system and therapist registration. Their duties include user creation, system monitoring through an admin dashboard. These users are expected to have a moderate to advanced set of technical expertise and knowledge of system security and data privacy principles.
- **Therapists:** They are the primary users of the system. They are responsible for reviewing patient profiles, analyzing behavioral data, starting AI-based batch analysis, interpreting explainable reports, and providing approvals for therapy-related insights. They are expected to have moderate computer literacy and familiarity with digital clinical or analytical tools, but no advanced technical expertise is required.
- **Caregivers:** Responsible for the daily collection of behavioral data, recording and uploading standardized action videos, required for analysis and therapy suggestions. The interface for the users will be designed to be highly usable and reducing complexity for these type of users.

3.2.3 Assumptions and Dependencies

The design and implementation of the Intelligent Behavior Analysis and Therapy Support System for Autism are based on the following assumptions and dependencies:

- Users will access the system using modern, up-to-date web browsers with cookies enabled.
- Stable and reliable internet connectivity is assumed for uploading behavioral videos, accessing dashboards, and performing AI-based batch analysis.
- The cloud-based video storage services used by the system will remain available, scalable, and capable of securely storing encrypted behavioral video data without data loss or corruption.
- The internal ML analysis service will remain operational and accessible through secured APIs to support behavioral pattern recognition and report generation.

- RAG-based service for generation of recommendations, will be available with acceptable latency and will comply with data privacy and security requirements.
- All behavioral data, videos, and patient-related information will only be processed after valid informed consent has been granted and recorded within the system.
- Therapists and caregivers will follow standardized behavioral action definitions and data collection protocols as specified by the system to ensure consistency and analytical precision.

3.3 Specific Requirements

3.3.1 Functional Requirements

Functional requirements describe the core services that the system must provide to support data collection, analysis, and clinical decision-making. Each requirement must be testable and traceable to ensure the system accurately supports behavior analysis.

3.3.1.1 Secure Multi-Role User Authentication

1. The user opens the login screen after accessing the system.
2. The system asks the user to enter email and password.
3. The system verifies the authenticity of the user.
4. The system identifies the user's role (Administrator, Therapist, or Caregiver) after successful verification.
5. The system grants access to the user based on the role.
6. An appropriate error message is displayed on failed authentication.

3.3.1.2 Role-Based Access Control

1. The system determines the permissions associated with the user's role after successful login.
2. The system restricts access to functionalities.
3. Administrators have access to system management and user administration modules.

4. Therapists have access to patient management, analysis, and reporting modules.
5. Caregivers are granted access only to assigned patient logging and data submission modules.

3.3.1.3 User Management (Admin)

1. The administrator accesses the admin dashboard after successful authentication.
2. The system displays a list of registered users and their assigned roles.
3. The Administrator creates new user accounts or modifies existing user details.
4. The system logs all administrative actions for audit purposes.

3.3.1.4 Manage Patient Profiles and Therapy Goals

1. The Therapist navigates to the patient management screen after successful authentication.
2. The system provides options to create, view and update patient profiles.
3. The Therapist enters patient demographic information and diagnostic details.
4. The Therapist defines measurable therapy goals for the patient,
5. The system validates the input data and securely stores the patient profile.

3.3.1.5 Structured Behavioral Incident Logging

1. The Caregiver logs into the system and accesses the behavior recording and uploading screen.
2. The system displays a list of patients assigned to the Caregiver.
3. The Caregiver selects a patient and uploads a behavioral video.
4. The system automatically captures the timestamp of the incident.
5. The system validates the input data and stores them in the database.
6. The logged data becomes available for Therapist review.

3.3.1.6 Guided Video Recording of Standardized Actions

1. The Caregiver navigates to the video recording module.
2. The system displays step-by-step guidance for recording standardized patient actions.
3. The Caregiver records videos for each of the predefined actions:
 - Arm swing (left and right)
 - Body swing
 - Chest expansion
 - Sing and clap
 - Drumming
 - Frog pose
 - Maracas shaking
 - Maracas forward shaking
 - Squat
 - Tree pose
 - Twist pose
4. The system validates video duration, format, and resolution.
5. The system associates each video with the corresponding patient and action type.

3.3.1.7 Video Collection and Secure Storage

1. The system encrypts the video data on successful upload.
2. The system stores the video in secure cloud storage.
3. The system attaches metadata including patient ID, action type, timestamp, and quality indicators.
4. The system organizes videos chronologically and by action type.
5. The system ensures that the videos recorded on different days are correctly associated for analysis.
6. Videos that fail quality checks are rejected with appropriate feedback.

3.3.1.8 Intelligent Behavioral Pattern Analysis

1. The Therapist initiates the analysis process for a selected patient.
2. The system verifies that all required action videos have been uploaded.
3. The system sends the video data to the Machine Learning analysis service.
4. The LSTM-based model analyzes behavioral patterns.
5. The system receives and stores the analysis results.

3.3.1.9 Explainable AI Report Generation

1. The machine learning service generates analysis reports for each video.
2. The report includes severity, ADOS score, restrictive and repetitive behavior score, social affect score and the joints that contribute to each of these predictions.
3. The system computes confidence scores and uncertainty indicators.
4. The system provides explanatory reasoning to support each AI prediction.
5. The generated reports are stored and made accessible to the Therapist.

3.3.1.10 Behavioral Analysis Report

1. The system aggregates individual video analysis reports.
2. A comprehensive report is generated for the patient.
3. The report summarizes behavioral patterns across all actions.
4. The report presents both technical and patient-friendly summaries.

3.3.1.11 Therapist Reports Review

1. The Therapist accesses the analysis and reporting module.
2. The system allows playback of recorded videos with slow motion and frame-by-frame playback.
3. The Therapist reviews reports.
4. The Therapist adds annotations or overrides AI conclusions.
5. The Therapist provides contextual clinical notes.
6. The Therapist marks reports as reviewed and approved using a digital signature.

3.3.1.12 Generation of personalized therapy recommendations

1. The system analyzes approved behavioral reports.
2. The system retrieves relevant clinical knowledge using a RAG framework.
3. LLM generates therapy support recommendations.
4. The recommendations are displayed on the Therapist's dashboard.
5. The Therapist reviews and validates the suggested interventions before use.

3.3.1.13 Report Generation and Export

1. The Therapist selects a patient or therapy period for reporting.
2. The system generates a summary report.
3. The report is exported in a secure, non-editable format
4. The system embeds metadata such as patient ID and generation date.
5. The exported report is available for printing or secure sharing.

3.3.1.14 Report Export

1. The Therapist initiates a data export request.
2. The system exports structured behavioral data in CSV format.
3. The system validates file integrity before download.

3.3.1.15 Informed Consent Management

1. The system presents an informed consent document to patients or guardians.
2. The document explains video recording, data storage, and AI analysis.
3. The system discloses AI capabilities, limitations, risks, and benefits.
4. Patients or guardians grant or revoke consent digitally.
5. The system records consent decisions with timestamps.
6. AI analysis functions are disabled immediately upon consent revocation.

3.3.2 Non-Functional Requirements

3.3.2.1 Performance

- The system is designed to display all data visualizations and results in 5 seconds for typical data loads.
- The Caregiver logging interface responds almost instantly to ensure minimal disruption during behavioral incident reporting.

3.3.2.2 Reliability

- The system operates consistently and reliably during normal use without unexpected failures.
- Data integrity is ensured through transactional database operations for all patient and behavioral records.
- The target system uptime is 90%, ensuring dependable access for users.

3.3.2.3 Scalability

- The system architecture is designed to scale efficiently to handle increasing volumes of behavioral data and a growing number of users.
- The system supports future expansion without requiring a redesign of the core components of the system.

3.3.2.4 Security

- The system enforces strict role-based access control to ensure that only authorized users can access sensitive functionalities and data.
- All sensitive information, including Protected Health Information (PHI), is encrypted during transmission using TLS/HTTPS and encrypted at rest within the database.
- The system follows standard security best practices to protect against common web vulnerabilities such as unauthorized access, data leakage, and injection attacks.
- User authentication and access-related activities are securely handled to maintain confidentiality and system integrity.

3.3.2.5 Maintainability

- The system utilizes a modular architecture with separated components such as the Frontend, Backend API, Database and Machine Learning Service.
- This approach simplifies troubleshooting, system updates, bug fixes, and the addition of new features.
- System documentation is maintained to support long-term maintainability.

3.3.2.6 Usability

- The system provides a user-friendly and intuitive interface to reduce user errors and training effort.
- The Caregiver logging screen is optimized to require minimal steps and interactions to record behavioral incidents efficiently.
- The Therapist dashboard presents analytical results in a clear and logical manner, enabling easier interpretation of behavior patterns.
- System feedback, prompts, and validation messages guide users through common tasks and decision-making processes.

3.3.2.7 Availability

- The system is available to authorized users 24/7, ensuring continuous access for caregivers and therapists.
- Backup mechanisms and fault-tolerant system components are implemented to ensure continued access in the event of partial system failure.
- The system is designed to quickly restore services after downtime, maintaining data consistency and service continuity.

3.3.2.8 Compatibility

- The system is compatible with all major modern web browsers, such as Google Chrome, Mozilla Firefox, and Microsoft Edge.
- The user interface is responsive and adapts seamlessly to different screen sizes, allowing usage on desktops, laptops, tablets, and mobile devices.
- The system supports interoperability between frontend, backend, and machine learning components through standardized APIs and data formats.

3.3.3 Interface Requirements

The system shall primarily support user interaction through a secure, responsive **web-based interface** optimized for desktop access.

- **User Interface (UI):** Graphical interfaces incorporating input fields, structured forms, interactive charts, and clear system messages.
- **Software Interfaces (APIs):** The system requires integration with a database management system, and an internal API to pass behavioral data to the dedicated Machine Learning analysis service.
- **Hardware Interfaces:** No direct interaction with specialized hardware is required, but the system must function reliably across major mobile operating systems (iOS and Android) via a web browser.

3.3.4 Data Requirements

The system shall store data in a document-oriented NoSQL database (MongoDB) to support flexibility, scalability, and handling of clinical data.

- **Data Collections:** The system must include distinct collections for Users (Therapist/Caregiver), Patient Profiles, Defined Target Behaviors, and logged Behavioral Incidents.
- **Validation:** Data validation shall be enforced at the input layer (UI) and the application layer (API) for all fields to ensure accuracy (e.g., checking data types, enforcing length limits, and validating dates).
- **Integrity:** Relationships between data entities must be maintained using appropriate unique identifiers and reference fields to prevent orphaned records. The system shall ensure referential consistency through application level checks to prevent orphaned records.

Chapter 4

System Design

4.1 System Architecture

4.1.1 System Architecture

4.2 Use Case Diagrams

4.3 ER / Class Diagrams

4.4 UI Mockups or Wireframes

4.4.1 Login

The Login screen provides a secure access point for users of all roles. The interface is displayed in the Figure 5.1. It takes an email and password as input for secure access to the system. There is an option for therapist registration. The therapist can register from the login screen. If the user forgets the password, the interface also provides functionality to reset the password in a secure way.

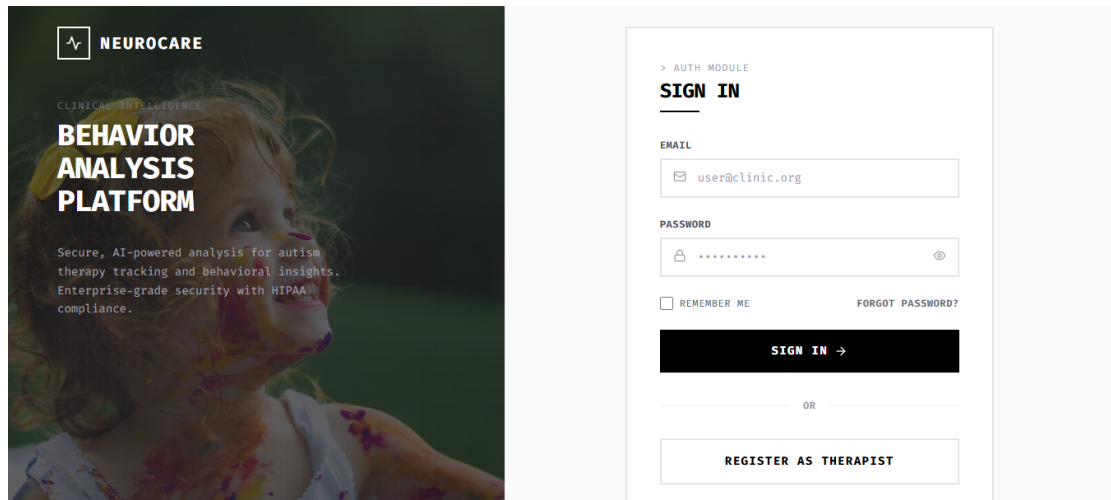


FIGURE 4.1: Login

4.4.2 Admin Dashboard

The admin dashboard is the interface built for administrators to manage the system, its users, approve therapists, and monitor the system. The interface, as shown in Figure 5.2, displays statistics, audit logs, system health, and therapist applications. The module provides user distributions, infrastructure metrics, and a refresh option to retrieve real-time updates.

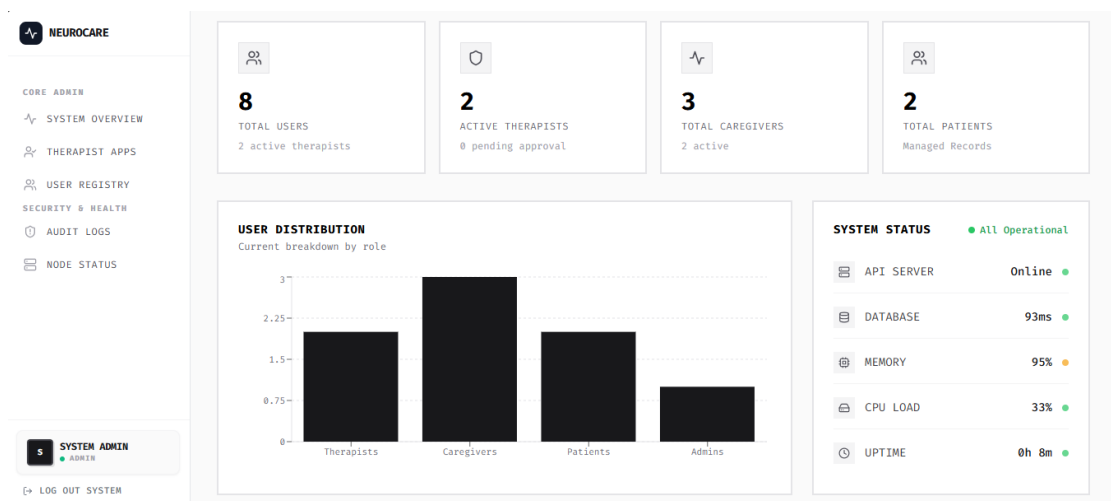


FIGURE 4.2: Admin Dashboard

4.4.3 Caregiver Dashboard

This interface, as shown in Figure 5.3, provides a caregiver focused dashboard for daily behavioral logging activities. It displays assigned patients, recording

tasks, and a guided video recording module. It also displays a section for behavior analysis and therapy suggestion reports.

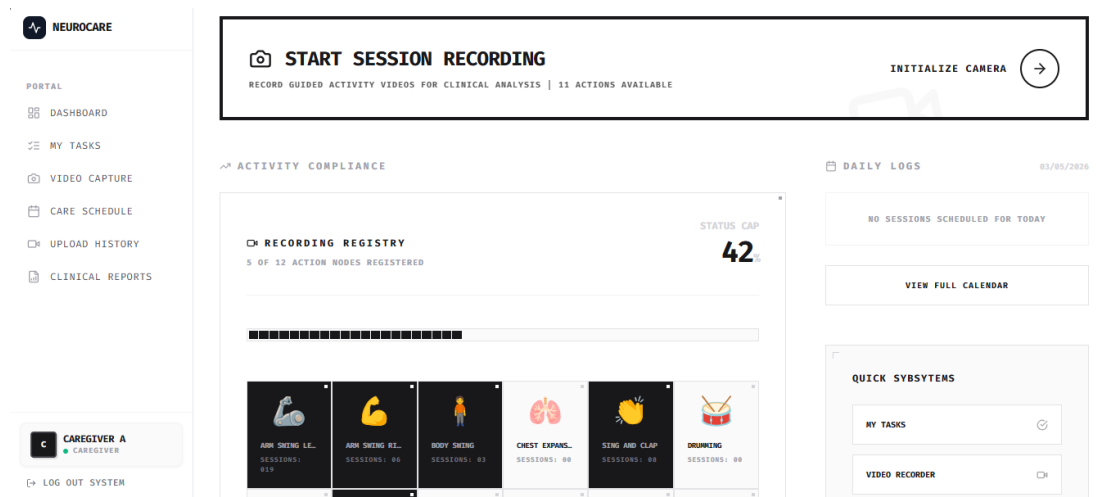


FIGURE 4.3: Caregiver Dashboard

4.4.4 Therapist Dashboard

The therapist dashboard as shown in Figure 5.4, provides an interface to the therapist to add patients, invite caregivers, assign patients to caregivers, assign tasks to caregivers, leverage behavior analysis module for patient video sessions, call therapy suggestions on multiple video sessions, and perform clinical workflow.

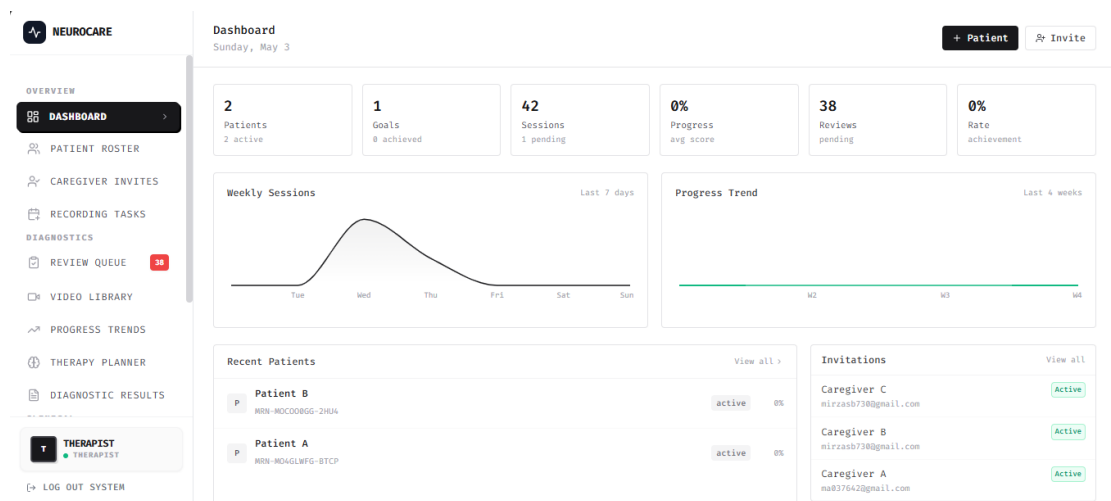


FIGURE 4.4: Therapist Dashboard

4.4.5 Guided Video Recording Screen

The interface as shown in Figure 5.5, displays the guided video recording module with clear instruction for caregiver to record video sessions. It provided step-by-step guidance to the caregiver to record the video of the patient and select appropriate action for video recording.

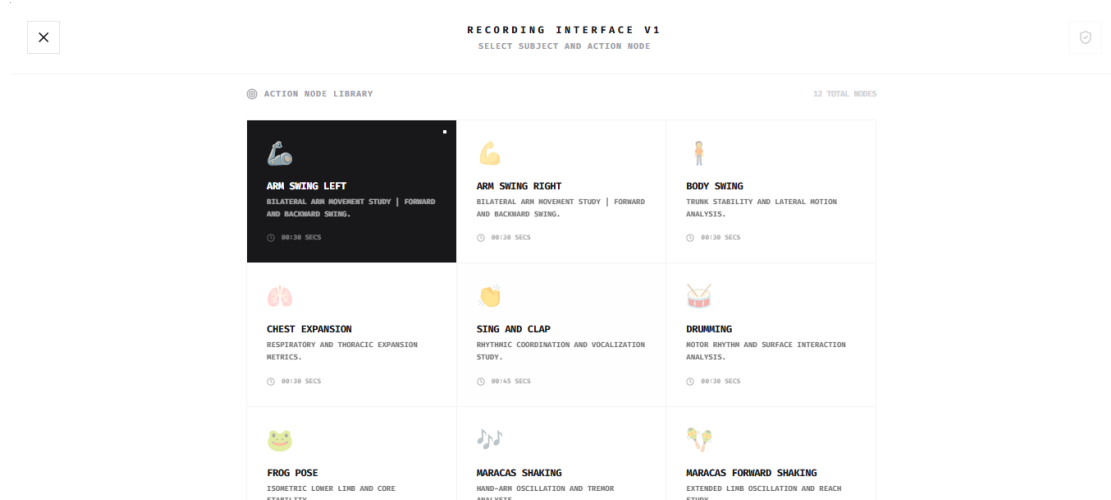


FIGURE 4.5: Guided Video Recording

4.4.6 Behavior Analysis Screen

The behavior analysis screen is the interface, as shown in Figure 5.6, which displays predictions from the behavior analysis engine. It displays the severity of autism, restrictive and repetitive behavior, social affect, and the ADOS comparison score based on the video analysis. It also provides video playback for the therapist to review model predictions and accept or reject the analysis. It displays detailed explainability and confidence metrics for diagnostic review, facilitating decision-making for the therapist.

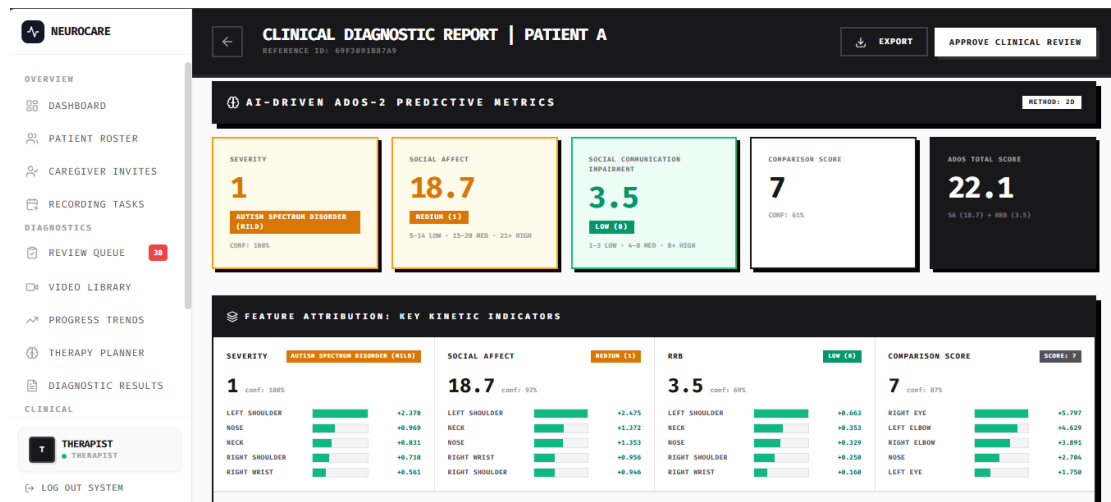


FIGURE 4.6: Behavior Analysis Screen

4.4.7 Therapy Suggestions Screen

The therapy suggestion interface, as shown in Figure 5.7, displays the RAG-based therapy recommendation. It allows the therapist to select the patient and the video sessions that he wants to use as context for therapy suggestions. The interface displays the AI-generated report, with the option for the therapist to edit the report and share it with the caregiver.

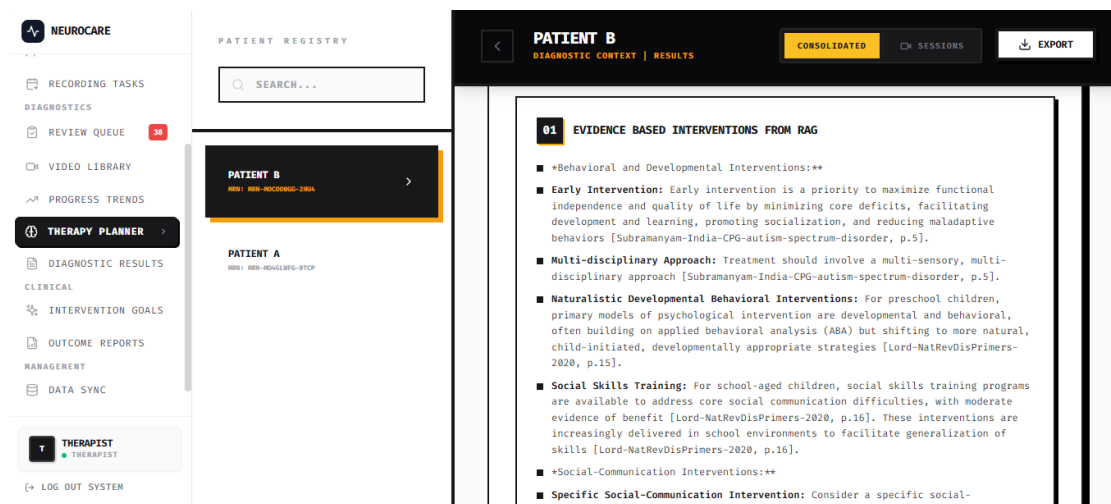


FIGURE 4.7: Therapy Suggestions Screen

Chapter 5

System Implementation

5.1 Development Tools and Technologies

5.1.1 Frontend: React

The frontend of this project was implemented using React with TypeScript because of its ability to use reusable components makes it easier to manage large interfaces. A component-based structure helped us to keep the frontend code organized and manageable. Vite was used as the build tool for the frontend because it provides very fast startup and refresh during development.

Benefits:

- Reusable UI components and cleaner code structure
- Fast development cycle with Vite
- Type safety with TypeScript reduces runtime errors

Implementation Summary:

The role-based screens and workflows were developed. Its routing was protected so that users can only access those pages allowed for their role. TanStack Query was used for data fetching and caching, and API communication was centralized through a shared client layer. This made state updates and error handling more consistent across the frontend.

5.1.2 Backend: Node.js with NestJS

The backend was developed using Node.js and NestJS because it provides a modular structure, strong support for TypeScript, and clear separation between controllers, services, and data models. This was useful for implementing multiple

modules such as authentication, users, patients, invitations, clinical workflows, and AI prediction and therapy module integration.

Benefits:

- Modular architecture for backend development
- Support for validation, guards, and dependency injection
- Easy maintenance and extension of features

Implementation Summary:

Role-based guards and authentication were applied to protected routes. The clinical module manages video sessions and review flow. A prediction module communicates with the AI service and stores results. Admin, user, and patient management were implemented as separate modules for clarity and maintainability.

5.1.3 Database: MongoDB

MongoDB was used as the primary database for this project because many records in the system are nested and evolve over time. For example, patient records, clinical session data, role-specific user metadata, and stored prediction outputs.

Benefits:

- Flexible schema
- Good fit for nested clinical data
- Strong integration with NestJS through Mongoose

Implementation Summary:

The collections in the database were designed around platform domains such as users, admins, patients, therapy goals, and video sessions. Indexing and query filtering were used for listing and retrieval operations. Prediction responses and clinical workflow states are stored for audits and follow-up reporting.

5.1.4 Authentication and Authorization: JWT and Role-Guards

The system issues a token generated using JSON Web Tokens (JWT) after successful login. The token is verified on each request by the protected APIs. Role-based access control measures have been implemented to enforce different permissions for admins, therapists and caregivers.

Benefits:

- Stateless authentication
- Easier integration
- Clear role-based access control

Implementation Summary:

The JWT token is generated after the user is successfully logged in. The token encapsulates the user identity and role information. The token is stored in the cookies and sent to the server for authorization of each request. Backend then validates the token and enforces role-based access control measures on endpoints. This ensures that user only interacts with those actions and data that are relevant to its role.

5.1.5 AI Service: Fast API

FastAPI was used to expose the endpoints for behavior analysis model and the RAG based therapy suggestion module. It was selected because of its lightweight nature and suitability for ML inference.

Benefits:

- Fast and lightweight for ML inference
- Easier integration due to Python ecosystem
- Separation of ML endpoints

Implementation Summary:

The AI service exposes two ML related endpoints of behavior analysis and therapy generation. The behavior analysis endpoint takes the video, perform pose estimation of MediaPipe for 2D and ROMP for 3D and then runs the machine learning models with those pose estimations as input and then responds with structured predictions and explainable outputs. The therapy generation takes the behavior analysis outputs as inputs and runs the RAG pipeline with curated clinical documents to generate evidence based therapy suggestions.

5.2 Description of Modules

5.2.1 Authentication and Authorization

This module handles all the secure access of the system and makes sure that the access control measures are setup. It makes sure that no person is able to access data that he is not authenticated for and is not able to perform any operation if he is not authorized for that operation.

Implementation Summary:

- JWT authentication is used for user identity validation for protected endpoints
- Role guards are used for role based access control.
- JWT token validation on each request prevents unauthorized operations.

5.2.2 User Management

The system contains a module of user management and provides different features and controls the access of features based on the role. **Implementation Summary:**

- Implemented flow of therapist registration, approval, login flow, caregiver invite and patient assignment and onboarding for both caregiver and therapist.
- Developed dashboard for admin, caregiver and therapist.
- Admin can remove and suspend users and approve therapists.

5.2.3 Invitation Module

The system has an invitation module which is responsible for inviting caregivers and onboarding therapists and caregivers.

Implementation Summary:

- Therapists can register from the login page. An email will be sent to their email where the onboarding process will be performed. Admin will verify the therapist.
- Therapist can invite caregivers after approval from admin and then assign patients to careg.

5.2.4 Patient Management

Patient management contains the patient creation, its assignment to the caregiver and record management for the clinical workflow.

Implementation Summary:

- Therapist can create patients and update their clinical data.
- Sessions, analysis and recommendations are recorded for the patients.

5.2.5 Video Session Management

This module handles the uploading, processing, encryption at rest and in transit, decryption, and storage of videos for behavioral logging of standardized actions.

Implementation Summary:

- The users will be properly guided on the video recording.
- Users can upload a video or record a video.
- The therapist can assign the recording task to the caregiver.
- Video recording, uploading, and storage will be managed by this module.
- End-to-end encryption of the patient video will be performed.

5.2.6 Behavior Analysis Module

This module performs behavior analysis on the patient video by processing pose estimation and then analyzing those estimations through a trained LSTM model.

Implementation Summary:

- The outputs of the pose estimation are used to create a feature vector. This feature vector is analyzed by an LSTM model.
- The LSTM model is trained with MMASD dataset.
- The module also explains the model predictions with the joints that are responsible for each model prediction.

5.2.7 Therapy Recommendation Module

This module is responsible for generating therapy suggestions based on the behavioral analysis.

Implementation Summary:

- Behavior predictions are used as structured query context.
- Relevant therapy evidence is retrieved from curated clinical knowledge sources.
- Recommendations are generated and returned in a review-friendly structured format.

- The behavior analysis are used as query context for generating therapy recommendations.
- RAG framework is used for evidence based therapy suggestions.
- Recommendations are returned in a structured format that is easy to review.

5.3 Key Implementation Details

- **Service Oriented Architecture:**

The system is designed with a separate frontend, backend, and AI service layer. This improved maintainability and allowed the team to develop each layer independently.

- **Frontend:**

The frontend of the system was built in React with reusable components and a centralized API caching layer with TanStack Query, which improves performance and refresh behavior across role based screens

- **Backend:**

The backend of the system was built in Nest which provides strong TypeScript support. This allowed the backend to be built in a modular way with reduced coupling.

- **AI Pipeline:**

The AI pipeline of the system involves pose estimation, temporal sequence modeling, and retrieval-augmented generation. This design helped analyze the behavior from the videos and then use this analysis to generate therapy recommendations with clinical evidence. Its APIs were exposed using FastAPI and the LSTM model was trained on MMASD dataset. The RAG pipeline was implemented using langchain.

- **Role-based Access Control:**

Access control measures were implemented based on the user role through JWT and Role Guards to ensure that each user can access only relevant features and data.

Chapter 6

System Testing and Evaluation

6.1 Testing Methodology

6.2 Frontend Testing

3.1 Objectives of Frontend Testing Validate UI functionality Ensure responsiveness and accessibility Verify user workflows 3.2 Functional Testing User registration/login Data input forms Report generation Therapy recommendation display Navigation testing 3.3 UI/UX Testing Responsiveness across devices Accessibility considerations Ease of use for therapists/parents 3.4 Frontend Performance Testing Page load time API response handling Rendering performance 3.5 Frontend Testing Results Screenshots/tables of results Issues identified and fixes

6.3 Backend Testing

4.1 Objectives of Backend Testing Validate API functionality Ensure database integrity Verify authentication and authorization 4.2 API Testing Request/response validation Error handling Status code verification 4.3 Database Testing Data storage and retrieval Data consistency Query performance 4.4 Security Testing Authentication testing Authorization testing Input validation Protection against common vulnerabilities

6.4 Bug Tracking and Resolution

6.5 Evaluation

6.5.1 Requirements Fulfillment

The system was evaluated against the original requirements of the system. The following table displays the key feature requirements of the system and their fulfillment.

TABLE 6.1: Requirement Fulfillment Status

ID	Requirement	Status	Evaluation Method
FR1	Secure multi-role user authentication and role-based access control	Fulfilled	Security testing
FR2	User management and administrative control with audit logging	Fulfilled	Security testing
FR3	Patient profile management and therapy goal definition	Fulfilled	Functional testing
FR4	Structured behavioral incident logging and guided video recording of standardized actions	Fulfilled	Usability testing
FR5	Secure video collection, encryption and storage	Fulfilled	Security testing
FR6	Intelligent behavioral pattern analysis	Fulfilled	ML performance metrics
FR7	Explainable AI report generation	Fulfilled	Model interpretability evaluation
FR8	Comprehensive behavioral report generation and therapist review workflow	Fulfilled	Functional testing
FR9	Therapy recommendation generation using RAG framework	Fulfilled	Retrieval relevance testing
FR10	Report generation, secure export, and sharing mechanisms	Fulfilled	File integrity testing and export validation
FR11	Informed consent management	Fulfilled	Compliance verification

6.5.2 Behavior Analysis Model Results

The behavior analysis model trained on the MMASD dataset produced the following evaluation metric results. The model produced excellent evaluation metrics

on training dataset. The following table displays the classification report on the severity and the comparison score attribute. The overall accuracy of the comparison score attribute is 0.97 and 0.94 respectively. The R-square evaluation metric is used to evaluate both restrictive and repetitive behavior and social affect. The R-square is 0.82 for RRB and 0.83 for social affect score.

TABLE 6.2: Classification Report for Comparison Score Attribute

	Precision	Recall	F1-Score
Score 6	0.92	0.92	0.92
Score 7	0.96	1.00	0.98
Score 8	0.92	0.88	0.90
Score 9	0.81	1.00	0.90
Score 10	0.97	0.95	0.96

TABLE 6.3: Classification Report for Severity Attribute

	Precision	Recall	F1-Score
Severity 2.0	0.92	0.92	0.92
Severity 3.0	0.96	1.00	0.98

Severity and ADOS comparison are classification tasks and show high precision and recall across most classes and strong overall accuracy. Social affect and restricted and repetitive behavior (RRB) are regression based tasks have 0.82 R-square evaluation metrics.

6.5.3 Therapy Suggestion Pipeline Results

6.5.4 System Strengths

- The behavior analysis engine analyzes the patient video sessions and acts as context for RAG-based therapy suggestions resulting in context-aware, evident therapy suggestions.
- Frontend, backend and AI service built separately in a modular architecture supporting scalability and modularity.
- All the AI outputs are generated with explainability for clinical review and interpretable for therapist.

6.5.5 Areas for Improvements

- Monitor the performance of the system in a real-world deployment environment.

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- Improve explainability by adding more explainable features like temporal segments, etc.
 - Improve the latency by optimizing the model.

Chapter 7

Conclusions and Future Direction

7.1 Key Achievements

- Behavior analysis model and RAG pipeline for therapy suggestions and implementation of explainability for model predictions due to the sensitivity of the domain.
- High classification metrics for classification tasks and average metrics for regression tasks.
- Integration of AI service with secure web platform.
- Development of an end-to-end platform containing video upload, encryption / decryption, pose estimation, model inference, retrieval, generation and results review.

7.2 Lessons Learned

- Ethical guidelines of systems for healthcare domain.
- Learned about the machine learning development life-cycle and the explainability in the model predictions.
- Clinician trust is increased when generating high grounding and relevance therapy suggestions.
- Modular architecture simplifies the development and deployment of the system.

7.3 Future Work

- Optimize the model performance to improve the model's latency and RAG response.
- Add more explainability to the behavior analysis engine for therapists.
- Expand the knowledge base of the RAG pipeline to improve the quality and authenticity of therapy suggestions.

7.4 Final Remarks

The delivered system shows that LSTM-based behavior analysis with explainability with retrieval-augmented generation with LLMs can generate evidence-based therapy recommendations for autism and can be of great value to a therapist as a support system. The behavior analysis engine shows great evaluation metrics for classification tasks but improvements can be made for regression tasks as this will increase the system's clinical utility. Collaboration with the clinical stakeholders will be essential for improvement of data, and the knowledge base and overall improvement of the system.

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